

ENTERPRISE CLOUD SPECIFICATION

Software Requirements Specification

Multi-Tenant University ERP & White-Label Web Platform

PROJECT NAME

UniFlow Enterprise ERP Web Platform

DOCUMENT VERSION

2.0.0 (Cloud-Native Edition)

COMPLIANCE STANDARDS

IEEE Std 830-1998 / ISO/IEC/IEEE 29148

TARGET AUDIENCE

University Executives, Technical Architects, Dev Team, QA

SECURITY & MULTITENANCY

PostgreSQL Row-Level Security (RLS) & Argon2id RBAC

LOCALIZATION

Bilingual Native (Arabic RTL / English LTR) & Dual Calendar

EXECUTIVE SUMMARY

UniFlow is an institutional-grade, multi-tenant cloud SaaS ERP engineered specifically for higher education. It unifies public white-label admissions, academic hierarchy setup, comprehensive student lifecycle management, and an immutable, double-entry financial accounting and cashiering engine.

1. Introduction

1.1 Purpose & Vision

This document specifies the software requirements for **UniFlow Enterprise ERP**, a modern, responsive, multi-tenant, cloud-native SaaS web platform. The platform is architected for commercial deployment across universities, colleges, and academies, delivering end-to-end institutional management.

The platform is anchored upon four core architectural pillars:

- **1. White-Label Public Portal & Admissions CMS:** Custom university branding, program discovery, and self-service student application portals.
- **2. Student Intake & Lifecycle Management:** Multi-step registration, biometric/photo capture, medical clearances, and status transitions.
- **3. Enterprise Double-Entry Accounting:** 5-tier Chart of Accounts, maker-checker voucher approvals, fee catalogs, deferred revenue recognition, cashiering POS, cheques, budgeting, fixed assets, and financial statements.
- **4. Modern Bilingual UI Engine:** Built on React, TypeScript, Tailwind CSS, and shadcn UI, offering zero-reload Arabic (RTL) and English (LTR) localization.

1.2 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition & Institutional Context
SaaS	Software as a Service — Multi-tenant cloud-hosted institutional architecture.
COA	Chart of Accounts (دليل الحسابات) — 5-tier standardized accounting tree.
GL	General Ledger (دفتر الأستاذ العام) — Core transactional journal store.
JV	Journal Voucher (قيد اليومية) — Double-entry financial balancing transaction.
RV / PV	Receipt Voucher (سند قبض) / Payment Voucher (سند صرف).
AR / AP	Accounts Receivable (Student Billing) / Accounts Payable (Vendor Billing).
RBAC	Role-Based Access Control governing granular operational permissions.
Maker-Checker	Separation of duties requiring voucher drafting and approval by distinct actors.
NBV	Net Book Value (القيمة الدفترية للأصل) — Asset cost minus accumulated depreciation.
Encumbrance	Budget commitment reserved upon Purchase Order approval prior to disbursement.
Functional Currency	The declared primary currency in which a tenant maintains its statutory ledgers.
Idempotency Key	Client-supplied token guaranteeing repeat API requests execute exactly once.
تفقيط (Tafqeet)	Algorithmic conversion of monetary amounts into formal written Arabic words on receipts.

2. Overall Description & System Architecture

2.1 Web Platform Architecture

UniFlow employs a distributed, high-concurrency web architecture comprising a Next.js/React frontend with Tailwind CSS and shadcn UI, REST/GraphQL API microservices, and a PostgreSQL database layer enforcing tenant isolation via Row-

Level Security (RLS). Redis provides low-latency idempotency token tracking and session management.

2.2 Core System Capabilities Matrix

Capability Area	Web Implementation Details	Security & Architectural Control
Authentication & IAM	Argon2id hashing, secure HTTP-only cookies, MFA for financial approvals, granular RBAC.	OWASP Top 10 Security Baseline
White-Label CMS	Per-tenant dynamic theme tokens, custom domains, HSL color palettes, typography selection.	Zero-Code Multi-Tenant Engine
Student Admission	Multi-step intake wizard, 4-part discrete names, photo capture, document checklist.	Client & Server Schema Validation
Registration & Billing	Atomic semester registration coupled with immediate double-entry GL posting in a single transaction.	ACID Database Transaction
Chart of Accounts	5-tier normalized tree with parent keys, account codes, normal balance flags, and bilingual titles.	Strict Level-5 Postability
General Ledger	Immutable transaction headers and lines with database-enforced debit=credit balancing triggers.	PostgreSQL Trigger Enforced
Voucher Approvals	4-stage workflow: Draft → Pending Review → Pending Approval → Posted. Full history logged.	Immutable State Transition Log
Cashiering POS	Multi-channel fee collection (Cash, Bank, Cheque, Gateway) with instant bilingual thermal/A4 receipts.	Idempotency Token Guaranteed
Cheque Management	Lifecycle state machine (Received → In Collection → Deposited → Cleared / Bounced) with automated GL entries.	State-Triggered Journaling
Fixed Assets	Automated straight-line depreciation engine, persisted schedule per asset, period-locked batch dispatch.	Idempotent Batch Runner
Budget & Encumbrance	Budget lines by fiscal year × account × cost center with real-time PO encumbrance reservation.	Real-Time Overrun Prevention
Financial Reports	Maintained period balance rollups ('account_period_balances') powering sub-100ms trial balance queries.	< 100ms Query Latency SLA

2.3 System Invariants & Core Design Principles

- 1. **Double-Entry Invariant:** Every journal voucher must satisfy $\sum \text{Debits} = \sum \text{Credits}$ in the functional currency, enforced by database triggers.
- 2. **Monetary Precision:** All financial values use `numeric(19,4)` to eliminate floating-point rounding errors.
- 3. **Idempotent Financial Execution:** Every state-changing financial request requires an idempotency key, preventing duplicate transactions on network retry.
- 4. **Financial Immutability:** Posted transactions cannot be edited or deleted. Corrections require explicit, linked reversal vouchers with audit comments.
- 5. **Multi-Tenant Isolation:** Tenant boundaries are enforced mathematically at the database layer via PostgreSQL Row-Level Security (RLS).

REQ-SOD-01: SEGREGATION OF DUTIES MANDATE

The platform strictly prohibits any single user role from combining: (a) Creating and Approving the same voucher; (b) Maintaining fee matrices and Authorizing discounts; (c) Editing vendor banking details and Approving vendor disbursements; (d) Posting journal vouchers and Unlocking closed fiscal periods.

3. Specific Functional Requirements

Module 1: White-Label Landing Page & CMS Engine

- **REQ-LP-01: Dynamic Branding:** University Admin configures institutional branding tokens (Bilingual Name, Logo, Favicon, Motto, HSL Color Palette, Typography) rendered dynamically across public pages.
- **REQ-LP-02: Hero Banner & CTA:** Customizable hero section with rich media backgrounds and direct portal action buttons.
- **REQ-LP-03: Program Explorer:** Public searchable catalog of Faculties, Degrees, study plans, career prospects, and published tuition schedules.
- **REQ-LP-04: Online Admissions:** Prospective student application wizard with certificate upload, choice ranking, tracking number generation, and online fee payment.
- **REQ-LP-05: News & Calendar:** CMS publishing academic calendars, semester milestones, and campus announcements.
- **REQ-LP-06: Campus Contact:** Map embed, campus directory, and inquiry routing forms.

Module 2: Academic Hierarchy & Configuration

- **REQ-AC-01: Faculties & Departments:** Organizational structure setup for university faculties, academic departments, and research centers.
- **REQ-AC-02: Degree Programs:** Degree level definitions (B.Sc., M.Sc., Ph.D., Diploma) with standard durations and credit requirements.
- **REQ-AC-03: Academic Calendar:** Academic years, semester terms (Fall, Spring, Summer), and batch cohort definitions.
- **REQ-AC-04: Versioned Fee Matrix:** Effective-dated fee matrix by Program × Batch × Admission Type × Nationality Category. Modifications generate an immutable version record.
- **REQ-AC-05: Admission Channels:** Master tables for national admission categories, expatriate quotas, and international fee rules.
- **REQ-AC-06: Cost Centers:** Multi-tier cost center catalog aligned with faculties and operational divisions.

Module 3: Student Admission & Digital Profile Management

- **REQ-ST-01: Profile Builder:** Structured 4-part Arabic and English names, unique University ID generation via configurable masks, demographics (DOB Gregorian/Hijri, National ID), and guardian profiles.
- **REQ-ST-02: Medical Fitness:** Health screening records capturing blood group, infectious disease checks, vaccination history, allergies, and doctor clearance verdicts (Fit / Unfit / Conditional).
- **REQ-ST-03: Normalized Search:** Sub-100ms debounced search with Arabic text normalisation (folding alef variants, taa marbuta, yaa/alef maqsura, and tatweel).
- **REQ-ST-04: Profile Audit History:** Field-level audit logging of all profile updates with actor ID, timestamp, and before/after values.
- **REQ-ST-05: Document Checklist:** Verification checklist for required admission credentials with expiry alerts for renewable documents.

Module 4: Semester Registration & Student Lifecycle

- **REQ-REG-01: Registration Workflow:** Student lookup → fee matrix resolution → discount/scholarship application → net tuition calculation.
- **REQ-REG-02: Atomic GL Posting:** Registration creation atomically posts a balanced double-entry transaction (Dr Student AR, Cr Unearned Fee Income) in the same database transaction.
- **REQ-REG-03: Cancellation & Reversal:** Registration cancellation posts a linked reversal voucher, preserving historical records.
- **REQ-REG-04: Program Transfers:** Program transfer workflow reversing prior program charges and posting new curriculum billing.
- **REQ-REG-05: Digital Registration Cards:** Generation of registration proof and student ID cards featuring a verifiable QR code.
- **REQ-REG-06: Registration Holds:** Automated holds (financial, academic, disciplinary) restricting registration until cleared.

Module 5: 5-Tier Chart of Accounts & General Ledger

- **REQ-FIN-01: 5-Tier Chart of Accounts:** (1) Major Class, (2) Group, (3) Sub-Group, (4) Parent Account, (5) Detail Account. Only Level-5 accounts accept direct postings.
- **REQ-FIN-02: Control Accounts:** Accounts designated as Student AR, Sponsor AR, or Vendor AP accept postings exclusively through sub-ledgers.
- **REQ-FIN-03: Multi-Line Journal Vouchers:** N-debit to N-credit journal entry grid with real-time balancing, multi-currency conversion, FX gain/loss handling, and file attachments.
- **REQ-FIN-04: Maker-Checker Approval:** 4-stage workflow: Draft → Pending Review → Pending Approval → Posted. High-value approvals require MFA re-authentication.
- **REQ-FIN-05: Immutable Voucher Reversals:** Posted vouchers cannot be modified. Adjustments require creating a linked reversal voucher with mandatory justification.
- **REQ-FIN-06: Document Sequences:** Per-tenant, per-fiscal-year gapless document numbering managed via transactional sequence allocators.

Module 6: Cashiering POS & Accounts Receivable

- **REQ-CSH-01: Multi-Channel Receipts:** Cashier terminal supporting Cash, Bank Transfer, Cheque, and Online Gateway payments with instant bilingual thermal/A4 receipts.
- **REQ-CSH-02: Installment Plans:** Configurable installment schedules per program (e.g. 50% registration, 25% midterm, 25% final) with automated SMS/Email reminders.
- **REQ-CSH-03: Payment Vouchers:** General disbursement vouchers for vendor invoices, refunds, and operating expenses with maker-checker approvals.
- **REQ-CSH-04: General Receipts:** Receipt vouchers for non-student revenues (grants, leases, donations, certificate fees).
- **REQ-CSH-05: Payment Gateway Engine:** Provider-agnostic payment gateway interface with webhook signature verification and automated reconciliation.
- **REQ-CSH-06: Receipt Cancellation:** Same-day receipt voiding by authorized supervisors generating automated linked reversal journals.

Module 7: Cheque Management & Clearing Pipeline

- **REQ-CHQ-01: Portfolio Tracking:** Registry of post-dated and on-demand cheques capturing number, drawee bank, due date, drawer name, and custody status.
- **REQ-CHQ-02: Clearing State Machine:** Transition pipeline: Received → Sent to Bank → Cleared (Dr Bank, Cr Cheques Rec.) / Bounced (Dr Student AR, Cr Cheques Rec.) / Cancelled.

- **REQ-CHQ-03: Bounced Cheque Handling:** Bounces reinstate student receivable balance, record bank refusal codes, and apply configurable penalty fees.

Module 8: Institutional Budgeting & Financial Control

- **REQ-BDG-01: Annual Fiscal Budget:** Budgets configured by Fiscal Year × Account × Cost Center with monthly/quarterly allocation curves.
- **REQ-BDG-02: Real-Time Budgetary Control:** Available = Allocated - Encumbered - Actual. System enforces configurable warning or hard-stop policies on overruns.
- **REQ-BDG-03: Variance Reporting:** Multi-dimensional reporting comparing Budgeted, Encumbered, Actual, Available, and Utilization %.

Module 9: Fixed Assets & Depreciation Engine

- **REQ-AST-01: Asset Registry:** Asset tracking (Lab equipment, IT, furniture, vehicles, buildings) with QR/Barcode, location, custodian, and historical cost.
- **REQ-AST-02: Depreciation Engine:** Automated straight-line depreciation engine generating persisted period schedules per asset.
- **REQ-AST-03: Automated Batch Journals:** Period-end automated batch posting (Dr Depreciation Expense, Cr Accumulated Depreciation). Runs are idempotent and period-locked.
- **REQ-AST-04: Asset Disposal & Revaluation:** Asset retirement, scrap, or revaluation with automated gain/loss calculation.

Module 10: Financial Statements & Business Intelligence

- **REQ-RPT-01: Student Statement:** Chronological ledger statement of charges, payments, and running balance with receipt references.
- **REQ-RPT-02: Receivable Aging:** Aging buckets (<30, 31-60, 61-90, >90 days) aged from installment due dates, filterable by faculty/program.
- **REQ-RPT-03: Trial Balance:** Multi-level Trial Balance with Opening, Movement, and Closing columns sourced directly from period-balance aggregators.
- **REQ-RPT-04: Multi-Level Balance Sheet:** Institutional Balance Sheet structured from Level 1 summary to Level 4 detail.
- **REQ-RPT-05: Income Statement:** Operating revenues vs. expenses, net surplus/deficit, filterable by Cost Center with comparative prior-period analysis.
- **REQ-RPT-06: Sub-Ledger Reconciliation:** Automated verification ensuring Sub-Ledger totals match General Ledger Control Account balances exactly.
- **REQ-RPT-07: Export Engine:** Export to Excel (.xlsx), CSV, and shaped bilingual PDF with official university header and digital stamps.

Module 11: Multi-Tenancy & Platform Administration

- **REQ-ADM-01: Multi-Tenant Architecture:** PostgreSQL Row-Level Security (RLS) and application middleware enforcing strict tenant isolation.
- **REQ-ADM-02: Granular RBAC:** Configurable role permissions governing view, create, edit, approve, and reverse actions per module.
- **REQ-ADM-03: Tamper-Evident Audit Log:** Cryptographically hash-chained, append-only audit trail logging user, IP, timestamp, action, and JSON diffs.
- **REQ-ADM-04: Bilingual RTL/LTR Engine:** Zero-reload language switching, native RTL mirroring, dual Gregorian/Hijri calendars, and Arabic monetary amount spelling (تفقيط).
- **REQ-ADM-05: Subscription Licensing:** SaaS tier enforcement controlling active modules, student capacity, and administrative seats.

Module 12: Fiscal Calendar & Opening Balances

- **REQ-PER-01: Fiscal Periods:** Fiscal years divided into monthly periods with states: Future, Open, Closed, Permanently Closed.
- **REQ-PER-02: Posting Period Lock:** System rejects postings into Closed periods. Period close requires passing automated reconciliation checks.
- **REQ-PER-03: Opening Balances:** Structured opening balance entry requiring exact debit=credit and sub-ledger reconciliation prior to go-live.
- **REQ-PER-04: Year-End Close:** Closing revenue and expense accounts into Retained Earnings / Surplus with balance carry-forward.

Module 13: Fee Item Catalog & Revenue Recognition

- **REQ-FEE-01: Fee Catalog:** Catalog of billable fees (Tuition, Registration, Labs, Exams, ID, Hostel, Transport) with designated GL revenue accounts.
- **REQ-FEE-02: Deferred Revenue:** Tuition billing credits Unearned Fee Income, recognized into revenue across semester periods via scheduled recognition jobs.
- **REQ-FEE-03: Withdrawal Refunds:** Automated refund calculation based on withdrawal date schedules, reversing unearned income and raising accounts payable.
- **REQ-FEE-04: Credit Balances:** Maintenance of student credit balances resulting from overpayments, auto-applied to future registrations.

Module 14: Student Status Lifecycle

- **REQ-LIF-01: Status Model:** State machine: Applicant → Admitted → Active → Deferred / Suspended / Withdrawn / Dismissed / Transferred / Graduated → Alumni.
- **REQ-LIF-02: Transition Governance:** Status changes require recorded justification, supporting documentation, and authorized workflow sign-off.
- **REQ-LIF-03: Re-Admission:** Re-admission workflow restoring prior academic history and applying the currently active fee matrix.

Module 15: Sponsors & Scholarships

- **REQ-SPN-01: Sponsor Master:** Registry of corporate sponsors, government ministries, and embassies with contract parameters and billing addresses.
- **REQ-SPN-02: Split Funding:** Automated billing split apportioning student charges between student self-pay and sponsor AR accounts.
- **REQ-SPN-03: Sponsor Invoicing:** Consolidated billing per sponsor per cycle with automated aging and payment matching.
- **REQ-SPN-04: Scholarship Governance:** Scholarship programs with allocated budgets, eligibility rules, approval thresholds, and foregone revenue exposure reports.

Module 16: Procurement & Accounts Payable with Encumbrance

- **REQ-PRC-01: Vendor Master:** Vendor directory, bank accounts, tax numbers, payment terms, and assigned AP control accounts.
- **REQ-PRC-02: Procure-to-Pay Workflow:** Requisition → Purchase Order (creates Encumbrance) → Goods Receipt → Vendor Invoice → Payment Voucher.
- **REQ-PRC-03: Encumbrance Accounting:** Automated commitment against budget lines upon PO approval, converting to accrued liability on goods receipt.
- **REQ-PRC-04: Three-Way Match:** Automated three-way matching among Purchase Order, Goods Receipt, and Vendor Invoice within configurable tolerances.

- **REQ-PRC-05: AP Aging & Payment Run:** Vendor invoice aging and batch payment proposal generator.

Module 17: Admissions Capacity & Committee Workflow

- **REQ-ADM-CAP-01: Seat Quota Management:** Live seat quotas per Program × Batch × Admission Channel preventing over-enrollment.
- **REQ-ADM-CAP-02: Automated Eligibility:** Screening of secondary certificate scores, required subjects, and age/nationality constraints.
- **REQ-ADM-CAP-03: Committee Review:** Committee scoring portal with candidate ranking and structured decision recording (Accept, Conditional, Waitlist, Reject).
- **REQ-ADM-CAP-04: Offers & Deposits:** Automated bilingual offer letter generation with expiration deadlines and online seat deposit collection.
- **REQ-ADM-CAP-05: Duplicate Detection:** Candidate matching on national ID, passport, and normalized Arabic name to prevent duplicate applications.
- **REQ-ADM-CAP-06: Bulk Intake Import:** Bulk import engine for centrally-allocated admissions rosters with pre-commit dry-run validation.

Module 18: Academic Records & Course Management (Phase 7-8 Roadmap)

- **REQ-ACD-01: Course Catalog:** Course master with codes, bilingual titles, credit/contact hours, and prerequisite/co-requisite trees.
- **REQ-ACD-02: Timetable Scheduling:** Section scheduling with instructor assignment, room capacity, and timetable clash detection.
- **REQ-ACD-03: Course Registration:** Student course enrollment with prerequisite verification, credit load limits, and automated financial hold blocking.
- **REQ-ACD-04: Attendance Tracking:** Session attendance tracking enforcing regional 75% examination eligibility rules with automated risk alerts.
- **REQ-ACD-05: Assessment & GPA:** Multi-component grade entry, approval workflow, GPA/cGPA calculation, and academic standing evaluation.
- **REQ-ACD-06: Examination Management:** Exam hall seating allocation, invigilator schedules, results publishing, and fee-bearing re-sit workflows.
- **REQ-ACD-07: Transcripts & Degree Audit:** Official bilingual transcripts with QR verification, degree audit validation, and graduation clearance.

4. Data Architecture & Database Schema

4.1 Database-Enforced Invariants

Invariant Rule	Database Enforcement Mechanism
Double-Entry Balance	Deferred PostgreSQL constraint trigger enforcing $\text{SUM}(\text{debits}) = \text{SUM}(\text{credits})$ in functional currency per voucher header.
Gapless Document Numbering	Transactional sequence table with <code>UNIQUE (tenant_id, fiscal_year_id, voucher_type, voucher_no)</code> .
Closed Period Lock	Database trigger on <code>transaction_headers</code> validating <code>fiscal_periods.status = 'Open'</code> prior to insert.
Strict Level-5 Postability	Foreign key check constraint ensuring postings only reference accounts where <code>is_postable = TRUE</code> .

Invariant Rule	Database Enforcement Mechanism
Sub-Ledger Control Integrity	CHECK constraint ensuring control accounts always include a valid subledger_id.
Ledger Immutability	PostgreSQL revoked UPDATE/DELETE privileges on transaction_headers and transaction_lines.
Zero Precision Loss	All financial columns declared strictly as numeric(19,4).
Multi-Tenant Isolation	PostgreSQL Row-Level Security (RLS) policies applied to all tenant operational tables.
Single Reversal Guarantee	UNIQUE index on (reverses_id) WHERE reverses_id IS NOT NULL ensuring vouchers are reversed at most once.
Tamper-Evident Audit Trail	Cryptographic SHA-256 hash chaining linking each audit log entry to its predecessor (prev_hash).

5. Non-Functional Requirements

5.1 Performance & Concurrency

- **REQ-NFR-01: Page Load:** Initial application load < 1.5 seconds; client-side route transitions < 200 milliseconds.
- **REQ-NFR-02: Query Latency:** Debounced student and account searches resolve in < 100 ms at scale (> 100k students, > 1M journal lines).
- **REQ-NFR-03: High Concurrency:** Sustained throughput of 500+ concurrent cashiering and registration transactions with zero deadlocks.

5.2 Security & Integrity

- **REQ-NFR-04: Cryptographic Standards:** TLS 1.3 in transit; AES-256 encryption at rest for databases and document vaults.
- **REQ-NFR-05: Authentication:** Secure/HttpOnly/SameSite session cookies; Argon2id password hashing; MFA for financial approvals.
- **REQ-NFR-06: Financial Immutability:** Prohibition of hard deletes or silent edits. All corrections require explicit linked reversals.
- **REQ-NFR-07: Financial Idempotency:** Mandatory idempotency tokens on all state-changing financial APIs.
- **REQ-NFR-08: Injection Immunity:** Parameterized prepared statements and typed ORMs across all data access paths.

5.3 Regional Localization & UI Ergonomics

- **REQ-NFR-09: Bilingual RTL/LTR:** Native Arabic RTL with Cairo/Inter typography; correct Arabic PDF text shaping; algorithmic Arabic monetary spelling (تفقيط).
- **REQ-NFR-10: Dual Calendar:** Simultaneous Gregorian and Hijri calendar support across all student-facing and official documents.
- **REQ-NFR-11: Responsive Layouts:** Fluid layouts across desktop displays, tablets, and POS mobile screens.
- **REQ-NFR-12: Theming Tokens:** shadcn UI / Tailwind CSS tokens supporting Light/Dark modes and university brand palettes.
- **REQ-NFR-13: Accessibility:** WCAG 2.2 Level AA compliance across all public and student portal surfaces.

5.4 Availability & Operability

- **REQ-NFR-14: Backup & Recovery:** RPO ≤ 15 minutes; RTO ≤ 4 hours with automated restore drill verification.

- **REQ-NFR-15: Observability:** Structured JSON logging with request correlation IDs and real-time alerting on ledger variances.
- **REQ-NFR-16: Data Export:** Automated tenant data export in open formats (CSV/JSON/SQL) with configurable retention policies.

6. Institutional Tenant Onboarding & Initialization

UniFlow enforces a 4-stage onboarding protocol ensuring that institutional tenants go live with fully reconciled, mathematically verified operational ledgers:

- **Stage 1 — Chart of Accounts Setup:** Import from institutional CSV templates or seed from the pre-packaged Higher Education COA template. Validates 5-tier depth, normal balances, and control account flags.
- **Stage 2 — Master Data Initialization:** Import of faculties, degree programs, batch cohorts, fee items, fee schedules, cost centers, and active student rosters with duplicate checking.
- **Stage 3 — Opening Balances & Reconciliation Gate:** Entry of opening GL balances and per-student, per-sponsor, and per-vendor sub-ledger balances. Go-live is gated on zero variance between total debits/credits and sub-ledger totals.
- **Stage 4 — Go-Live & Operational Cutover:** User provisioning, RBAC role assignment, Segregation of Duties review, opening of the first fiscal year/periods, and cashier drawer allocation.

7. Strategic Product Roadmap & Future Extensions

Functional Area	Target Capability	Strategic Value & Impact
Cashier Shift Controls	Shift open/close with denomination counting, Z-reports, and petty cash imprest management.	Enhanced teller drawer security and cash handling governance.
Payroll & HR-Lite	Staff registry, grade pay scales, monthly payroll batch posting, and end-of-service accruals.	Unifies university human capital expenditure directly into the GL.
Offline PWA Terminal	Offline-tolerant cashier Progressive Web App queueing receipts with local cryptographic signing.	Guarantees cashier continuity during campus network or power outages.
Automated Communications	Multi-channel SMS, WhatsApp, and Email dunning pipeline for payment reminders and notices.	Accelerates student accounts receivable collection cycles.
Ancillary Billing	Hostel bed allocation, transportation route subscriptions, and library fee integration.	Consolidates campus service revenues into the central student ledger.
Advanced Analytics & BI	Cash flow forecasting, student enrollment funnel conversion, and regulatory statistical returns.	Delivers executive decision support for university boards and deans.
SaaS Platform Billing	Automated subscription metering, institutional invoice generation, and credit card / wire processing.	Automates commercial billing for multi-institution deployments.